Strings in C

- A **string in C** is an **array of characters ending with \0** (null terminator)
  `char str[] = "Hello"` → stored as `'H' 'e' 'l' 'l' 'o' '\0'`
- ⚠️ Always ensure enough space for `\0`!

- Use `<string.h>` for string functions:
  - `strlen(str)` – length (excludes `\0`)
  - `strcpy(dest, src)` – copy
  - `strcmp(str1, str2)` – compare
  - `strcat(dest, src)` – concatenate

- Use `<ctype.h>` for string functions:
  - `isdigit(int c)`
  - `isalpha(int c)`
  - `isspace(int c)`
  - `islower(int c)/isupper(int c)`

- **String** vs **String literal**:

Type	Example	Writable?
String literal	`char *s = "Hello"`	❌ read-only
String	`char s[] = "Hello"`	✅ writable

Converting `str` to `int`

Sample outputs

```
    45
The string "    45" corresponds to integer 45
rt45
The string "rt45" corresponds to integer 0
45yhu
The string "45yhu" corresponds to integer 45

The string "" corresponds to integer 0
 4 5
The string " 4 5" corresponds to integer 4
```

Converting str to int using ASCII values

	+0	+1	+2	+3	+4	+5	+6	+7
32		!	"	#	$	%	&	'
40	(	)	*	+	,	-	.	/
48	0	1	2	3	4	5	6	7
56	8	9	:	;	<	=	>	?
64	@	A	B	C	D	E	F	G
72	H	I	J	K	L	M	N	O
80	P	Q	R	S	T	U	V	W
88	X	Y	Z	[	\	]	^	_
96	`	a	b	c	d	e	f	g
104	h	i	j	k	l	m	n	o
112	p	q	r	s	t	u	v	w
120	x	y	z	{	|	}	~	

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72	H	I	J	K	L	M	N	O
80	P	Q	R	S	T	U	V	W
88	X	Y	Z	[	\	]	^	_
96	`	a	b	c	d	e	f	g
104	h	i	j	k	l	m	n	o
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120	x	y	z	{	|	}	~	

The ASCII code for the character ‘0’ is 48.

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The ASCII code for the character ‘0’ is 48.
The ASCII code for the character ‘3’ is 51.

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The ASCII code for the character '3' is 51.

⇒ To get the corresponding integer, always subtract 48.

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112	p	q	r	s	t	u	v	w
120	x	y	z	{	|	}	~	

The ASCII code for the character ‘0’ is 48. 48-48 = 0 ✓

The ASCII code for the character ‘3’ is 51. 51-48 = 3 ✓

⇒ To get the corresponding integer, always subtract 48.